INTERNSHIP PROJECT REPORT

Task 5

Ensemble Learning, Model Optimization & Real-World ML Pipeline

Name: CHANDI PRASAD PADHI

College: ITER

University: Siksha 'O' Anusandhan University

Branch: B.Tech (CSE)

Semester: 2nd Year / 4th Semester

Internship: MainCrafts Technology

Submission Date: 19th Aug 2026

Ensemble Learning and Model Optimization Objective:

To apply ensemble learning techniques for classification, compare model performance, use cross-validation for reliability, perform hyperparameter tuning, analyze feature importance, and demonstrate an end-to-end machine learning workflow.

Key Concepts:

- Ensemble Learning — combining multiple models to improve predictive performance.
- Random Forest — an ensemble of decision trees using bagging and feature randomness.
- Gradient Boosting — sequential learners that focus on correcting previous errors.
- Hyperparameter Tuning — optimization using GridSearchCV.
- Cross-Validation — estimating model reliability across multiple folds.
- Feature Importance — interpreting influential input features.
- ML Pipeline — connecting preprocessing, training, validation, tuning and selection.

Workflow:

Data Preparation → Model Training → Ensemble Models → Cross-Validation → Hyperparameter Tuning → Feature Importance → Model Comparison → Final Model Selection.

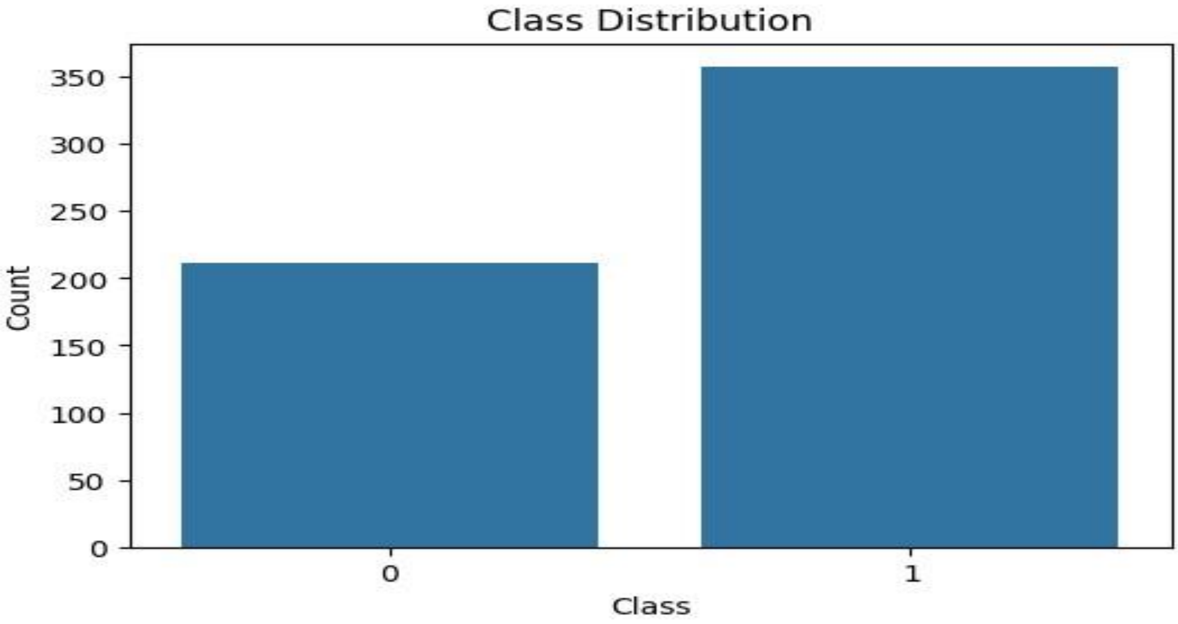


Figure 1. Task 5 visualization from the notebook.

Random Forest, Gradient Boosting and Validation Random Forest:

Random Forest combines multiple decision trees to reduce variance and improve generalization compared with a single decision tree. It is one of the core ensemble models covered in Task 5.

Gradient Boosting:

Gradient Boosting builds weak learners sequentially, with each new learner attempting to correct errors made by earlier learners. This provides a powerful boosting-based ensemble approach.

Cross-Validation:

K-fold cross-validation evaluates a model on multiple train-validation splits, providing a more reliable estimate of generalization performance.

Hyperparameter Optimization:

GridSearchCV systematically evaluates combinations of hyperparameters using cross-validation and selects the configuration that gives the best validation performance.

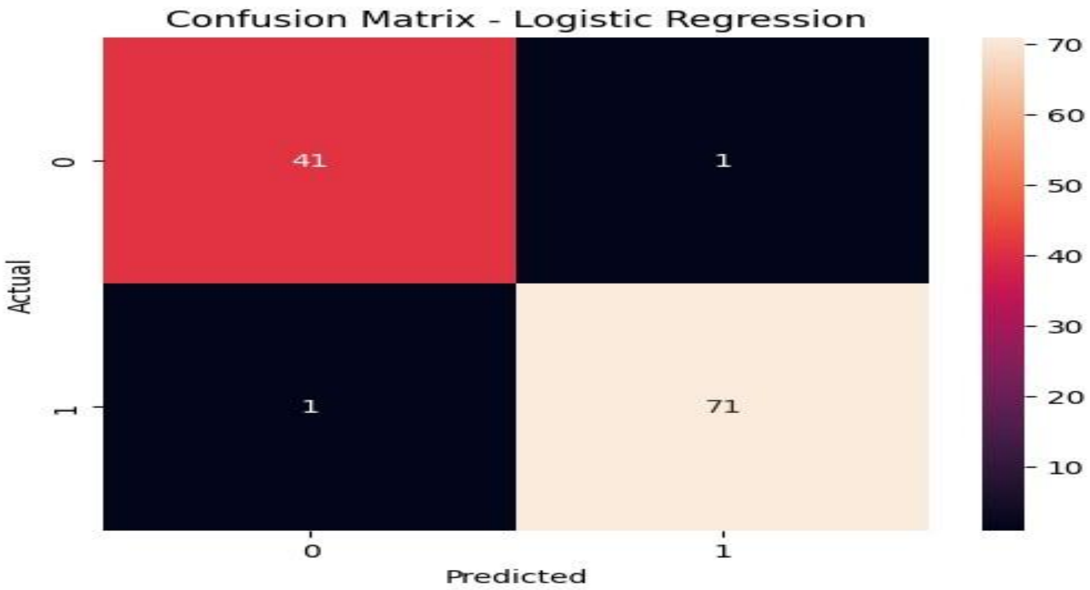


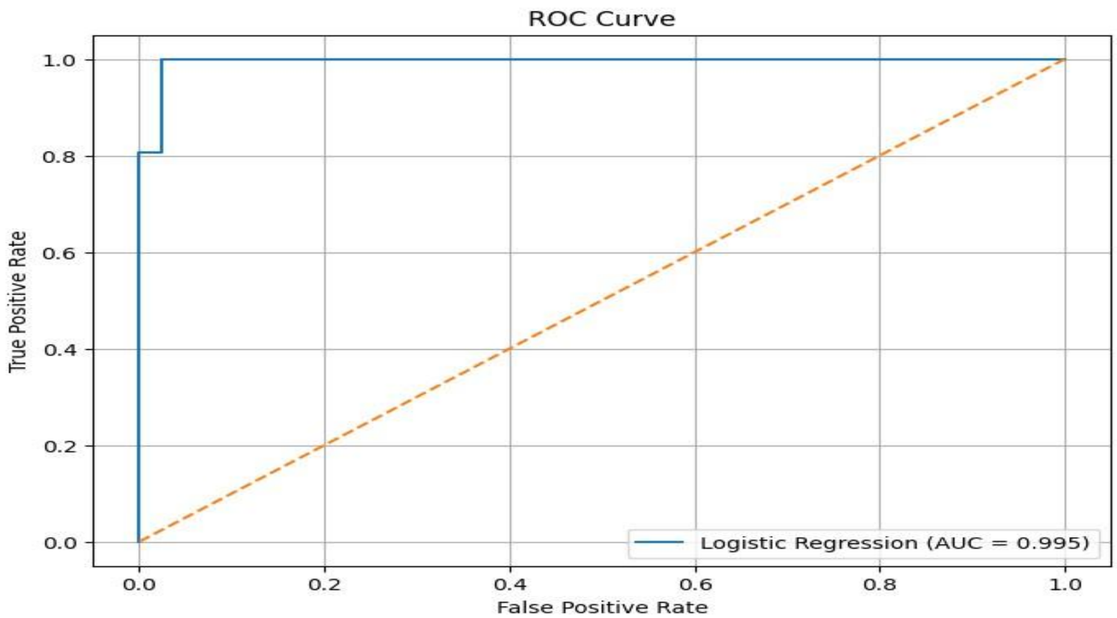
Figure 2. Model evaluation/validation result from the notebook.

Feature Importance, Model Comparison and Conclusion Feature Importance:

Feature importance ranks input variables according to their contribution to an ensemble model's predictions. This provides an interpretable view of which features influence the model most.

Model Comparison:

The Task 5 workflow compares ensemble and baseline models using appropriate classification metrics and cross-validation performance. The final model is selected using the notebook's evaluation results and generalization performance.



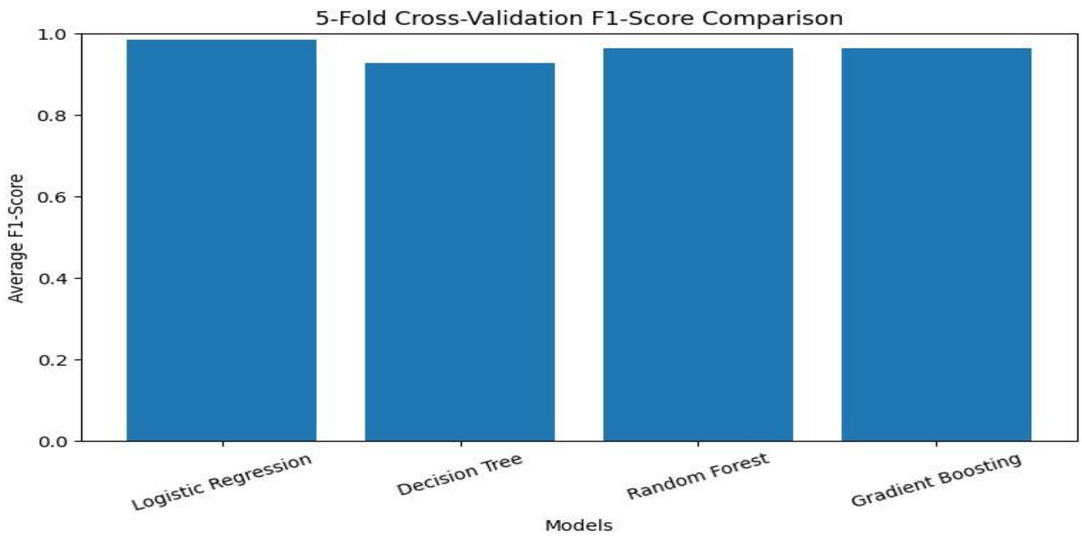


Figure 4. Additional Task 5 visualization.

Conclusion:

Task 5 extends machine learning from individual classifiers to ensemble learning and model optimization. Random Forest and Gradient Boosting demonstrate ensemble methods, while cross-validation and GridSearchCV improve reliability and tuning. Feature importance adds interpretability, completing an end-to-end workflow from preprocessing through model selection.